

**A REPORT
ON
AI RESUME ANALYZER**

Submitted by

Mr.TEJAS T O- 20241CSI0008

Under the guidance of,

Mr.Jinesh V N

Mr.Muthugurunathan G

in partial fulfillment for the award of the

degree of

BACHELOR OF TECHNOLOGY

IN

**COMPUTER SCIENCE AND ENGINEERING
(COMPUTER ENGINEERING)**

AT



PRESIDENCY UNIVERSITY

BENGALURU

AUGUST 2026

PRESIDENCY UNIVERSITY
PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND
ENGINEERING



CERTIFICATE

This is to certify that the report of **CSE7000 – Internship** on “**AI GPU Program**” being submitted by “**TEJAS T O**” bearing roll number “**20241CSI0008**” in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in **Computer Science and Information Technology (Computer Science and Engineering)** is a bonafide work carried out under our supervision.

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PRESIDENCY UNIVERSITY

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DECLARATION

I hereby declare that the work, which is being presented in the report entitled “**AI GPU PROGRAM**” in partial fulfillment for the award of Degree of **Bachelor of Technology in Computer Science and Information Technology(Computer Science and Engineering)**, is a record of my own investigations carried under the guidance of **Mr.Jinesh V N** and **Mr.Muthugurunathan G**, Presidency School of Computer Science and Engineering, Presidency University, Bengaluru.

I have not submitted the matter presented in this report anywhere for the award of any other Degree.

TEJAS T O
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INTERNSHIP COMPLETION CERTIFICATE



SCHOOL OF ARTIFICIAL INTELLIGENCE AND ADVANCED COMPUTING PRESIDENCY AI Centre of Excellence (Accelerated by NVIDIA)

Ref No: PU/PSAIAC/SIP/2026/20241CSI0008

Date: June 24th, 2026

To,
Mr. / Ms. **TEJAS T O**
Roll No: **20241CSI0008**
Program / Specialization: **CSI - Computer Science & Information Technology**

Semester: **5th Semester**, Undergraduate
Presidency University, Bengaluru

Subject: Offer of Admission and Guidelines for the Intensive AI GPU Summer Internship Program (Summer 2026)

Dear **TEJAS T O**,

Following the rigorous evaluation of your academic standing, motivation statement, we are pleased to offer you formal admission to the intensive, 2-week **AI GPU Summer Internship Program (Summer 2026)** at Presidency University. This elite training track is organized by the Presidency School of Artificial Intelligence and Advanced Computing in direct collaboration with our **NVIDIA Accelerated AI Centre of Excellence**. By accepting this offer, you will join a select cohort of students training on world-class computational architecture.

1. Individualized Training Schedule, Venue & Fee Allocations

Your program assignment contains specific technical parameters mapped to your ledger profile. You are required to attend sessions strictly within your allocated block:

Parameter	Individualized Allocation Detail
Assigned Cohort Phase Dates	Slot 3 : July 13 to July 25 (10 days)
Capstone Project Demo & Review	25th July 2026
Daily Time Slot Allocation	Slot B: 01:00 PM - 04:00 PM
Assigned Laboratory Venue	Lab (DGL04) - D-Block GPU Computing Lab Cluster
Registration Fee Category	Rs. 500/- Paid
Program Structural Volume	3 Hours/day over 10 Instructional Days (Total: 30 Contact Hours)
Academic Credit Load	2 Transferable Academic Credits

ACKNOWLEDGEMENTS

First of all, I indebted to the **GOD ALMIGHTY** for giving me an opportunity to excel in our efforts to complete this internship on time.

I express sincere thanks to our respected Dean **Dr. Duraipandian N**, Presidency School of Computer Science and Engineering & Presidency School of Information Science, Presidency University for getting us permission to undergo the internship.

I express heartfelt gratitude to our beloved **Dr.Pallavi R**, Head of the Department, Presidency School of Computer Science and Engineering, Presidency University, for rendering timely help in completing this internship successfully.

I am greatly indebted to my reviewers **Mr.Jinesh V N** and **Mr.Muthugurunathan G**, Presidency School of Computer Science and Engineering, Presidency University for their inspirational guidance, and valuable suggestions and for providing a chance to express technical capabilities in every respect for the completion of the internship work.

I would like to convey gratitude and heartfelt thanks to the Internship Coordinator **Dr. Bhuvaneshwari Patil** and Program Internship Coordinator **Mr.Teja Sirapu**. I also thank my family and friends for the strong support and inspiration they have provided us in bringing out this internship.

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ABSTRACT

The AI Resume Analyzer is an intelligent application designed to automate and simplify the process of resume screening and job matching. In traditional recruitment processes, recruiters need to manually review a large number of resumes, which can be time-consuming and inefficient. This project addresses this problem by using Artificial Intelligence and Natural Language Processing (NLP) techniques to analyze resumes and compare them with specific job descriptions.

The system allows users to upload a resume in PDF format and enter a relevant job description. The uploaded resume is processed using PyPDF2 to extract its textual content. Important skills, keywords, and relevant information are then identified from the extracted text. The system uses TF-IDF (Term Frequency–Inverse Document Frequency) to represent important terms in the resume and job description. Cosine similarity is applied to measure the similarity between them and generate a matching score.

The application is developed using Python and Streamlit, providing a simple and user-friendly interface. It helps users understand how well a resume matches a particular job requirement and identifies the skills that are relevant to the position. The system can be useful for both recruiters and job seekers by reducing manual screening effort and providing quick insights into candidate suitability.

The proposed system improves the efficiency of the recruitment process by providing automated resume analysis, skill identification, job-description matching, and candidate evaluation. It can reduce the time required for initial resume screening and support more consistent evaluation of candidates.

Overall, the AI Resume Analyzer demonstrates how AI and NLP can be applied to recruitment and career management to make the hiring process faster, easier, and more effective.

Introduction

Recruitment is an important process for organizations to identify and select suitable candidates for available job positions. In traditional recruitment, recruiters manually examine a large number of resumes to determine whether candidates possess the required skills, qualifications, and experience. This process can be time-consuming, repetitive, and difficult when dealing with a large number of applications. Therefore, there is a need for an automated system that can analyze resumes efficiently and assist in identifying suitable candidates.

The **AI Resume Analyzer** is a software application developed to automate the initial stage of resume screening and job matching. The system uses **Artificial Intelligence (AI), Natural Language Processing (NLP), and Machine Learning techniques** to extract useful information from resumes and compare it with the requirements mentioned in a job description. It helps determine how closely a candidate's resume matches a particular job role. In this system, the user can upload a resume in **PDF format** and provide a job description. The application extracts the textual content from the resume using **PyPDF2** and processes the extracted information. Relevant skills, keywords, and other important terms are identified from the resume. The system then uses **TF-IDF (Term Frequency–Inverse Document Frequency)** to represent the important terms and **Cosine Similarity** to measure the similarity between the resume and job description.

The application is developed using **Python and Streamlit**, which provides an interactive and user-friendly interface. The system generates a matching score that indicates the suitability of the resume for the given job description. This allows users to quickly understand whether their resume contains the skills and keywords required for a particular position.

The AI Resume Analyzer can be useful for **recruiters, organizations, students, and job seekers**. It reduces the time and effort required for initial resume screening and provides a consistent approach to resume evaluation. Overall, the project demonstrates how AI and NLP technologies can be applied to improve the recruitment process and support faster, more efficient, and data-driven candidate selection.

Objectives

- 1. Automate Resume Screening – Reduce the time required to manually review resumes.**
- 2. Extract Resume Information – Identify important details, skills, and keywords from resumes.**
- 3. Analyze Candidate Skills – Detect relevant skills mentioned in the candidate's resume.**
- 4. Process Job Descriptions – Analyze job requirements to identify expected skills and keywords.**
- 5. Perform Job Matching – Compare resume content with the given job description.**
- 6. Calculate Matching Score – Generate a score based on the similarity between the resume and job requirements.**
- 7. Improve Screening Efficiency – Make the initial candidate evaluation faster and easier.**
- 8. Reduce Manual Effort – Minimize repetitive work involved in resume evaluation.**
- 9. Support Job Seekers – Help candidates understand how well their resume fits a specific job.**
- 10. Assist Recruitment Decisions – Provide useful insights for better and faster candidate selection.**

WORK PROGRESS

The **AI Resume Analyzer** project developed through the following stages:

- **Requirement Analysis** – Identified the need for an automated resume screening system.
- **Data Collection** – Prepared resume and job-description data for analysis.
- **Resume Processing** – Implemented PDF text extraction using PyPDF2.
- **Skill Extraction** – Identified relevant skills and keywords from resumes.
- **Text Preprocessing** – Cleaned and prepared the extracted text for analysis.
- **TF-IDF Implementation** – Converted resume and job-description text into numerical vectors.
- **Similarity Calculation** – Used cosine similarity to compare resumes with job descriptions.
- **Matching Score** – Generated a score representing the suitability of the resume.
- **User Interface** – Developed an interactive interface using Streamlit.

RESULTS

The developed **AI Resume Analyzer** successfully automates the initial process of resume screening and job matching. The system was tested using sample resumes and different job descriptions to evaluate its ability to extract relevant information and determine candidate suitability. The application successfully accepts resumes in PDF format and extracts their textual content for further processing.

The system effectively identifies important **skills, keywords, and relevant information** from the uploaded resume. It also processes the provided job description and identifies the terms and requirements associated with the particular job role. This allows the system to compare the candidate's profile with the expected requirements of the position.

Using **TF-IDF (Term Frequency–Inverse Document Frequency)**, the important terms in the resume and job description are converted into numerical representations. **Cosine similarity** is then applied to determine the level of similarity between the two texts. Based on this similarity, the system generates a matching score that indicates how closely the candidate's resume corresponds to the given job description.

The developed Streamlit interface provides a simple and user-friendly environment where users can upload a resume, enter a job description, and obtain the analysis results. The system reduces the need for manual comparison and provides results quickly, making the initial screening process more efficient.

The testing results demonstrate that the system can successfully perform **resume text extraction, skill identification, keyword matching, job-description comparison, and candidate suitability evaluation**. It can also

help job seekers understand whether their resumes contain the skills and keywords required for a particular position.

Overall, the project demonstrates the practical application of **Artificial Intelligence, Natural Language Processing, TF-IDF, and cosine similarity** in the recruitment domain. The developed system provides a faster and more consistent approach to initial resume screening and can serve as a useful tool for both recruiters and job seekers.

CONCLUSION

The AI Resume Analyzer successfully demonstrates the application of Artificial Intelligence and Natural Language Processing in the recruitment and career management domain. The main objective of the project was to develop an automated system that could analyze resumes and compare them with job descriptions, reducing the time and effort involved in traditional manual resume screening.

The developed system allows users to upload resumes in PDF format and provide a job description for analysis. The resume text is extracted using PyPDF2 and processed to identify important information, skills, and keywords. The system then applies TF-IDF (Term Frequency–Inverse Document Frequency) to represent the important terms in both the resume and job description. Cosine similarity is used to determine the level of similarity between the two texts and generate a matching score.

The project provides a simple and interactive interface using Streamlit, making it easy for users to upload resumes and obtain analysis results. The system successfully performs important functions such as resume text extraction, skill identification, keyword analysis, job-description matching, and candidate suitability evaluation. These features help reduce repetitive manual work and make the initial screening process faster and more efficient.

The AI Resume Analyzer can be useful for both recruiters and job seekers. Recruiters can use the system to quickly evaluate multiple resumes and identify candidates whose profiles are more closely related to specific job requirements. Job seekers can use it to understand how well their resumes match a particular job and identify relevant skills or keywords that may need

improvement.

Although the current system focuses mainly on text-based resume analysis and similarity scoring, it provides a strong foundation for further development. Future improvements can include advanced machine learning models, improved skill extraction, experience and education analysis, resume recommendations, multiple resume comparison, and integration with online job portals.

Overall, the project demonstrates that AI-based resume analysis can make the recruitment process faster, more efficient, and less dependent on manual screening. It successfully combines Python, NLP, TF-IDF, cosine similarity, PyPDF2, and Streamlit to create a practical and user-friendly resume analysis solution.

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